MSG No. 131.439 $P4'_2/mmc'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `mmm`.

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `mmm`.

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `mmm`.

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 4: Wyckoff site: `2d`, site symmetry: `mmm`.

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[9,10,11,12,13,14,15,16]`

Table 5: Wyckoff site: `2e`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,4,6,7,11,12,13,14]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,3,5,8,9,10,15,16]`

Table 6: Wyckoff site: `2f`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,4,6,7,11,12,13,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,5,8,9,10,15,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `2mm`.

No.	position	mapping
1	$[0, 0, z]$	`[1,4,6,7]`
2	$[0, 0, -z]$	`[2,3,5,8]`

continued ...

Table 7

No.	position	mapping
3	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[9,10,15,16]
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[11,12,13,14]

Table 8: Wyckoff site: 4h, site symmetry: 2mm.

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[1,4,6,7]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[2,3,5,8]
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[9,10,15,16]
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[11,12,13,14]

Table 9: Wyckoff site: 4i, site symmetry: 2mm.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,4,6,7]
2	$\left[0,\ \frac{1}{2},\ -z\right]$	[2,3,5,8]
3	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[9,10,15,16]
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[11,12,13,14]

Table 10: Wyckoff site: 4j, site symmetry: m2m.

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,2,7,8]
2	$[-x,\ 0,\ 0]$	[3,4,5,6]
3	$\left[0,\ x,\ \frac{1}{2}\right]$	[9,11,14,16]
4	$\left[0,\ -x,\ \frac{1}{2}\right]$	[10,12,13,15]

Table 11: Wyckoff site: 4k, site symmetry: m2m.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[1,2,7,8]
2	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[3,4,5,6]
3	$\left[\frac{1}{2},\ x,\ 0\right]$	[9,11,14,16]
4	$\left[\frac{1}{2},\ -x,\ 0\right]$	[10,12,13,15]

Table 12: Wyckoff site: 4l, site symmetry: m2m.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	[1,2,7,8]
2	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[3,4,5,6]
3	$[0,\ x,\ 0]$	[9,11,14,16]

continued ...

Table 12

No.	position	mapping
4	$[0, -x, 0]$	[10,12,13,15]

Table 13: Wyckoff site: 4m, site symmetry: m2m.

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,2,7,8]
2	$\left[-x, \frac{1}{2}, 0\right]$	[3,4,5,6]
3	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[9,11,14,16]
4	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[10,12,13,15]

Table 14: Wyckoff site: 8n, site symmetry: ..2'

No.	position	mapping
1	$\left[x, x, \frac{1}{4}\right]$	[1,11]
2	$\left[x, -x, \frac{3}{4}\right]$	[2,10]
3	$\left[-x, x, \frac{3}{4}\right]$	[3,9]
4	$\left[-x, -x, \frac{1}{4}\right]$	[4,12]
5	$\left[-x, -x, \frac{3}{4}\right]$	[5,15]
6	$\left[-x, x, \frac{1}{4}\right]$	[6,14]
7	$\left[x, -x, \frac{1}{4}\right]$	[7,13]
8	$\left[x, x, \frac{3}{4}\right]$	[8,16]

Table 15: Wyckoff site: 8o, site symmetry: .m.

No.	position	mapping
1	$[0, y, z]$	[1,6]
2	$[0, -y, -z]$	[2,5]
3	$[0, y, -z]$	[3,8]
4	$[0, -y, z]$	[4,7]
5	$\left[-y, 0, z + \frac{1}{2}\right]$	[9,15]
6	$\left[y, 0, z + \frac{1}{2}\right]$	[10,16]
7	$\left[y, 0, \frac{1}{2} - z\right]$	[11,13]
8	$\left[-y, 0, \frac{1}{2} - z\right]$	[12,14]

Table 16: Wyckoff site: 8p, site symmetry: .m.

No.	position	mapping
1	$\left[\frac{1}{2}, y, z\right]$	[1,6]
2	$\left[\frac{1}{2}, -y, -z\right]$	[2,5]
3	$\left[\frac{1}{2}, y, -z\right]$	[3,8]
4	$\left[\frac{1}{2}, -y, z\right]$	[4,7]
5	$\left[-y, \frac{1}{2}, z + \frac{1}{2}\right]$	[9,15]

continued ...

Table 16

No.	position	mapping
6	$\left[y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[10,16]
7	$\left[y,\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[11,13]
8	$\left[-y,\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[12,14]

Table 17: Wyckoff site: 8q, site symmetry: m..

No.	position	mapping
1	$[x,\, y,\, 0]$	[1,8]
2	$[x,\, -y,\, 0]$	[2,7]
3	$[-x,\, y,\, 0]$	[3,6]
4	$[-x,\, -y,\, 0]$	[4,5]
5	$\left[-y,\, x,\, \frac{1}{2}\right]$	[9,14]
6	$\left[y,\, -x,\, \frac{1}{2}\right]$	[10,13]
7	$\left[y,\, x,\, \frac{1}{2}\right]$	[11,16]
8	$\left[-y,\, -x,\, \frac{1}{2}\right]$	[12,15]

Table 18: Wyckoff site: 16r, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[x,\, -y,\, -z]$	[2]
3	$[-x,\, y,\, -z]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$[-x,\, -y,\, -z]$	[5]
6	$[-x,\, y,\, z]$	[6]
7	$[x,\, -y,\, z]$	[7]
8	$[x,\, y,\, -z]$	[8]
9	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	[9]
10	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	[10]
11	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[11]
12	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[12]
13	$\left[y,\, -x,\, \frac{1}{2}-z\right]$	[13]
14	$\left[-y,\, x,\, \frac{1}{2}-z\right]$	[14]
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[16]